Thermodynamics and Statistical physics
2nd Test

Name: .....

Neptun code: .....

I. THEORY QUESTIONS

1. State Boltzmann's definition of the entropy. Use it to explain why the entropy should be increasing in time in an isolated system that is initially not in equilibrium. (2p)

2. Write down the canonical distribution for the probabilities of microstates in a system at temperature T. Define the partition function. How is it related to the Helmholtz free energy? (3p)

3. Write down the Maxwell-Boltzmann distribution for the probability of observing a particle with speed v in an ideal gas. Sketch a plot of the distribution and denote the mean speed, the root mean square speed and the most probable speed. (3p)

4. Define the diffusion constant for the motion of a particle in a fluid and express it in terms of the mean free path and the scattering time. (2p)

5. Draw the P-V diagrams corresponding to the van der Waals equation of state both above and below the critical temperature T_c. In the second case, denote the points corresponding the the liquid and gas phases (3p)

6. What is the distinction between a first and a second order phase transition? (2p)

7. What is Planck's statement of the 3rd law? What does it imply about the behavior of heat capacity at low temperatures? (2p)

II. PROBLEMS TO SOLVE

1. A 10 liter container contains nitrogen gas (molar mass 28 g/mol) at a pressure of 10^5 Pa and at a temperature of 300 K. Answer the following questions using the Maxwell-Boltzmann distribution.

   (a) Calculate the most probable speed of the gas molecules.

   (b) Determine the number of gas molecules with a velocity of 100 m/s or less.

   Hint: You may need the following approximation:

   $$e^{ax} \approx 1 + ax, \quad \text{if} \quad x \ll 1.$$

2. Two problems on phase transitions. The following data can be used.

   Specific heat values:

   - ice: 2.1 kJ/(kg K)
   - water: 4.2 kJ/(kg K)
   - copper: 390 J/(kg K)

   Latent heat of fusion: 334 kJ/kg

   (a) A piece of copper of mass $m = 90$ g at a temperature $T_1 = 90$ °C was placed in a calorimeter in which ice of mass 50 g was at a temperature -3 °C. Find the entropy increment of the piece of copper by the moment the thermal equilibrium is reached.

   (b) Find the increment of the ice melting temperature in the vicinity of 0 °C when the pressure is increased by $\Delta p = 1.00$ atm. The specific volume of ice exceeds that of water by $\Delta V = 0.091$ cm^3/g.

3. Consider a system of N distinguishable particles, each with three energy levels:

   $$\epsilon_0 = 0,$$
   $$\epsilon_1 = \epsilon,$$
   $$\epsilon_2 = 2\epsilon.$$

   (a) Calculate the single-particle partition function Z_1.

   (b) Derive the Helmholtz free energy F and entropy S of the system.

   (c) Using the partition function, find the internal energy U.

   (d) Verify that $F = U - TS$.